

Simple Machines — Lesson 2

ICSE Class 6 Physics · 10 MCQs · 1 mark each · Name: _____ Date: _____

Syllabus: Pulley and its mechanical advantage, wheel and axle, inclined plane, wedge, screw, care of machines.

Instructions: Each question has one correct option. Tick or circle the letter of the best answer. Attempt all questions.

1. A pulley is a wheel with a grooved rim over which passes a:

- | | |
|---------------|-------------------|
| A. chain only | B. wooden plank |
| C. metal rod | D. rope or string |

2. The ideal mechanical advantage of a single fixed pulley is:

- | | |
|--------|------|
| A. 2 | B. 1 |
| C. 0.5 | D. 4 |

3. A single fixed pulley is useful because it:

- | | |
|--|------------------------------|
| A. changes the direction of the effort | B. removes the load's weight |
| C. creates energy | D. doubles the effort |

4. The ideal mechanical advantage of a single movable pulley is:

- | | |
|--------|------|
| A. 0.5 | B. 3 |
| C. 1 | D. 2 |

5. Which of these works as a wheel and axle?

- | | |
|---------------------|--------------|
| A. a steering wheel | B. a knife |
| C. a ramp | D. a see-saw |

6. In a wheel and axle, the mechanical advantage equals:

- | | |
|-------------------------------------|-------------------------------------|
| A. load x effort | B. length / height |
| C. radius of wheel / radius of axle | D. radius of axle / radius of wheel |

7. A plank 5 m long is used to raise a load to a height of 1 m. The ideal MA of this inclined plane is:

- | | |
|------|--------|
| A. 5 | B. 0.2 |
| C. 4 | D. 6 |

8. Which of the following is an example of a wedge?

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|---------------------|----------------------|
| A. a steering wheel | B. a flagpole pulley |
| C. an axe | D. a see-saw |

9. The spiral thread of a screw is an example of:

- | | |
|---------------------|--|
| A. a wheel and axle | B. an inclined plane wrapped around a cylinder |
| C. a pulley | D. a lever |

10. To keep a machine working efficiently we should:

- | | |
|-------------------|-----------------------------------|
| A. let it rust | B. oil its moving parts regularly |
| C. never clean it | D. increase the friction |

Answer Key

1. D

2. B

3. A

4. D

5. A

6. C

7. A

8. C

9. B

10. B